```
R version 4.5.2 (2025-10-31) -- "[Not] Part in a Rumble"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin20

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  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[R.app GUI 1.82 (8556) aarch64-apple-darwin20]

[Workspace restored from /Users/admin/.RData]
[History restored from /Users/admin/.Rapp.history]

> genes <- read.csv("/Users/admin/Desktop/bee_assignment_gene_expression/genes_with_motif.txt", header=FALSE)
> library(clusterProfiler)
clusterProfiler v4.18.4 Learn more at https://yulab-smu.top/contribution-knowledge-mining/

Please cite:

S Xu, E Hu, Y Cai, Z Xie, X Luo, L Zhan, W Tang, Q Wang, B Liu, R Wang, W Xie, T Wu, L Xie, G Yu.
Using clusterProfiler to characterize multiomics data. Nature Protocols. 2024, 19(11):3292-3320

Attaching package: ‘clusterProfiler’

The following object is masked from ‘package:stats’:

    filter

Warning message:
package ‘clusterProfiler’ was built under R version 4.5.3
> library(org.Hs.eg.db)
Loading required package: AnnotationDbi
Loading required package: stats4
Loading required package: BiocGenerics
Loading required package: generics

Attaching package: ‘generics’

The following objects are masked from ‘package:base’:

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff, setequal, union


Attaching package: ‘BiocGenerics’

The following objects are masked from ‘package:stats’:

    IQR, mad, sd, var, xtabs

The following objects are masked from ‘package:base’:

    anyDuplicated, aperm, append, as.data.frame, basename, cbind, colnames, dirname, do.call,
    duplicated, eval, evalq, Filter, Find, get, grep, grepl, is.unsorted, lapply, Map, mapply, match,
    mget, order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank, rbind, Reduce, rownames,
    sapply, saveRDS, table, tapply, unique, unsplit, which.max, which.min

Loading required package: Biobase
Welcome to Bioconductor

    Vignettes contain introductory material; view with 'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.

Loading required package: IRanges
Loading required package: S4Vectors
```

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Attaching package: 'S4Vectors'

The following object is masked from 'package:clusterProfiler':

    rename

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname


Attaching package: 'IRanges'

The following object is masked from 'package:clusterProfiler':

    slice


Attaching package: 'AnnotationDbi'

The following object is masked from 'package:clusterProfiler':

    select


Warning message:
package 'S4Vectors' was built under R version 4.5.3
> gene_ids <- bitr(genes, fromType = "ENSEBL", toType = "ENTREZID", OrgDb=org.Hs.eg.db)
Error in bitr(genes, fromType = "ENSEBL", toType = "ENTREZID", OrgDb = org.Hs.eg.db) :
  'fromType' should be one of ACCNUM, ALIAS, ENSEMBL, ENSEMBLPROT, ENSEMBLTRANS, ENTREZID, ENZYME, EVIDENCE,
EVIDENCEALL, GENENAME, GENETYPE, GO, GOALL, IPI, MAP, OMIM, ONTOLOGY, ONTOLOGYALL, PATH, PFAM, PMID, PROSITE,
REFSEQ, SYMBOL, UCSCKG, UNIPROT.
> gene_ids <- bitr(genes, fromType = "ENSEMBL", toType = "ENTREZID", OrgDb=org.Hs.eg.db)
Error in .testForValidKeys(x, keys, keytype, fks) :
  None of the keys entered are valid keys for 'ENSEMBL'. Please use the keys method to see a listing of valid
arguments.
> gene_ids <- bitr(
+   genes,
+   fromType = "ENSEMBL",
+   toType = "ENTREZID",
+   OrgDb = org.Hs.eg.db
+ )
Error in .testForValidKeys(x, keys, keytype, fks) :
  None of the keys entered are valid keys for 'ENSEMBL'. Please use the keys method to see a listing of valid
arguments.
> head(genes)
               V1
1 ENSG00000292338
2 ENSG00000274231
3 ENSG00000224485
4 ENSG00000237616
5 ENSG00000176679
6 ENSG00000092377
> gene_ids <- bitr(
+   genes$V1,
+   fromType = "ENSEMBL",
+   toType = "ENTREZID",
+   OrgDb = org.Hs.eg.db
+ )
'select()' returned 1:many mapping between keys and columns
Warning message:
In bitr(genes$V1, fromType = "ENSEMBL", toType = "ENTREZID", OrgDb = org.Hs.eg.db) :
  17.28% of input gene IDs are fail to map...
> head(gene_ids)
          ENSEMBL  ENTREZID
1 ENSG00000292338    207063
4 ENSG00000237616 106480718
5 ENSG00000176679     90655
6 ENSG00000092377     90665
7 ENSG00000205726      6453
9 ENSG00000223078 106481961
> ego <- enrichGO(gene = gene_ids$ENTREZID, OrgDb = org.Hs.eg.db, ont = "BP")
```

```
> ekegg <- enrichKEGG()
Reading KEGG annotation online: "https://rest.kegg.jp/link/hsa/pathway"...
Reading KEGG annotation online: "https://rest.kegg.jp/list/pathway/hsa"...
Error in enrichKEGG() : argument "gene" is missing, with no default
gene
> ekegg <- enrichKEGG(gene=gene_ids$ENTREZID, orgnaism = "hsa")
Error in enrichKEGG(gene = gene_ids$ENTREZID, orgnaism = "hsa") :
  unused argument (orgnaism = "hsa")
> ekegg <- enrichKEGG(gene=gene_ids$ENTREZID, orgnanism = "hsa")
Error in enrichKEGG(gene = gene_ids$ENTREZID, orgnanism = "hsa") :
  unused argument (orgnanism = "hsa")
> ekegg <- enrichKEGG(gene=gene_ids$ENTREZID, organism = "hsa")
kegg_category.rda is not found, download it online...
> ekegg <- enrichKEGG(gene=gene_ids$ENTREZID, organism = "hsa")
> library(enrichplot)
enrichplot v1.30.5 Learn more at https://yulab-smu.top/contribution-knowledge-mining/

Please cite:

Guangchuang Yu, Li-Gen Wang, and Qing-Yu He. ChIPseeker: an R/Bioconductor package for ChIP peak
annotation, comparison and visualization. Bioinformatics. 2015, 31(14):2382-2383
> dotplot(ego)
> dotplot(ekegg)
> dotplot(ego)
> dotplot(ekegg)
> png("ego.png", width=800, height=600)
> ego_plot <- dotplot(ego)
> print(ego_plot)
> dev.off()
null device
          1
> png("/Users/admin/Desktop/bee_assignment_gene_expression/ekegg.png", width=800, height=600)
> ekegg_plot <- dotplot(ekegg)
> print(ekegg_plot)
> dev.off()
null device
          1
>
```